Rishabh Mahajan

+91-8427975008 • **rishabh7552@outlook.com** • **LinkedIn** • **GitHub Profile**

SUMMARY

AI/ML Engineer with 7 years shipping production software, owning the end-to-end model lifecycle — data curation, fine-tuning, evaluation, serving, and drift monitoring. Work spans classical ML, deep learning internals, and LLM systems built both in production and from scratch, with deep domain expertise in logistics, customs, and financial-crime compliance.

SKILLS SUMMARY

Machine Learning: scikit-learn, statsmodels, XGBoost, LightGBM, CatBoost, ensemble methods, GLMs, feature engineering, cross-validation, cost-based threshold optimization, probability calibration, SHAP interpretability

Deep Learning: PyTorch, Transformers, GPT pretraining from scratch, Vision Transformers (ViT), LoRA / PEFT, KV cache, vLLM, distributed training (DDP, Ray)

LLM & GenAI: Supervised Fine-Tuning (SFT), Hugging Face ecosystem, RAG architectures, LangChain, LangGraph, FAISS vector search, embeddings, structured output enforcement, LLM evaluation

MLOps & Serving: MLflow, DVC, Weights & Biases, Docker, FastAPI, GitHub Actions + CML, CI/CD pipelines, GCP Cloud Run, model governance, PSI drift detection, Streamlit

Data & Backend: Python, SQL, pandas, NumPy, Matplotlib, high-volume document & tabular data pipelines, automated schema validation, REST API integration

Cloud & Platforms: Azure OpenAI / AI Foundry, Azure Document Intelligence, GCP (Compute Engine, Cloud Run), Linux, Git, VS Code

WORK EXPERIENCE

DSV (formerly DB Schenker) – Gurugram, India

May 2023 – Present

AI / Software Engineer

- **Fine-tuned Qwen 2.5 7B** for bill-of-lading field extraction across 60 carriers, resolving unstructured carrier-specific layouts that template-based OCR could not parse. Document-level accuracy — every field in a document extracted correctly — rose from **~75–80% with prompted GPT-4/Claude to ~90%** on a strictly held-out test set of ~4,000 documents.
- **Built the training data curation pipeline** over **~36,000 masked documents** (~600 per carrier), fusing Azure Document Intelligence layout markdown with Kofax KTA OCR text as dual model input to achieve measurable accuracy gains on degraded scans.
- **Authored end-to-end training and inference code** — LoRA at rank 32 across all seven linear projections in `bf16` (~80M trainable parameters, 1.06% of base) — validating configuration on a carrier-level held-out split before retraining across all 60 carriers and packaging checkpoints, adapters, and inference services for Azure MLOps deployment.
- **Engineered deterministic schema and business-rule validation** in the inference layer with automated self-correction retries (up to 3 attempts) on failed extractions, **cutting manual review effort by 78%**.
- **Architected a parallel Azure Document Intelligence pipeline** using prebuilt and custom neural models for high-volume structured documents, processing **~5,000 documents/month at 96% field-extraction accuracy**.
- **Took a RAG assistant from POC to production** over automation execution reports for business stakeholders — a LangGraph state machine over FAISS retrieval and Azure OpenAI, served via FastAPI on Azure containers to **~2,000 users across 700+ automation processes**.
- **Engineered automated backend logistics workflows and enterprise integrations** — real-time TMS currency rate updates, customs clearance processing, shipment creation, route-deviation alerting, and automated credit-risk and SDLC governance pipelines.

Infosys – Mysore, India | Amsterdam, Netherlands

Jan 2019 – May 2023

Technology Analyst (promoted from Senior Systems Engineer / Trainee)

- **Delivered automated KYC verification and risk-analysis workflows** for the Detecting Financial Crime (DFC) program across 12+ client processes at **99% accuracy**, then optimized them to cut execution time ~15% and maintenance effort ~25%.
- **Built Python utilities for automated post-deployment checks**, cutting deployment timelines 20%; founded and led the automation security chapter, mentoring **15+ engineers** with zero high-severity incidents.

TECHNICAL PROJECTS (PRODUCTION MLOPS & DEEP LEARNING INTERNALS)

nanoGPT from Scratch – LLM Pretraining on FineWeb-Edu [GitHub] PyTorch • A100 80GB • RoPE / SwiGLU

- **Executed three controlled 124M-parameter pretraining runs** at matched compute (14,190 steps × 524,288 tokens = **7.44B tokens each**) on a single A100 80GB GPU, isolating transformer block design and data freshness against a byte-identical validation set.
- **Implemented LLaMA-style architectural upgrades** (RMSNorm, RoPE, SwiGLU) that cut validation loss from 3.093 to 3.055 with 0.8M fewer parameters, matching baseline final loss in **~24% less compute**; a single epoch of fresh tokens reached **3.046** (best overall).
- **Diagnosed the train–val loss gap as a bias–variance signal** — 0.06 under three epochs of repeated tokens vs. 0.02 on fresh data, separating memorization from generalization under weight decay 0.1 and gradient clipping. Scored **0.3056 HellaSwag `acc_norm` vs. 0.2955 for OpenAI's GPT-2 124M** at **56% MFU** with custom KV-cache inference.

Calibrated Clinical Risk Scoring & End-to-End MLOps System on GCP (Heart Disease Classification) [Live Project] [GitHub] FastAPI • Docker • Cloud Run • MLflow

- **Pooled four multi-institutional UCI clinical sites** (918 rows, 55% prevalence) into parallel scikit-learn prediction and statsmodels inference pipelines; reached **0.891 test ROC-AUC** (0.869 out-of-fold) at 0.131 Brier score and 0.984 recall, with a cost-derived threshold of 0.20 (missed diagnosis priced at 4× a false alarm) backed by reliability calibration curves.
- **Evaluated Leave-One-Site-Out (LOSO) cross-validation** as the honest generalization number (~0.70 AUC at two held-out sites vs. 0.891 on pooled splits). A tuned XGBoost challenger tied under paired bootstrap but ranked last on LOSO (0.787 vs. 0.797), its edge traced to missingness acting as a site proxy, so the interpretable linear champion stayed in production.
- **Audited algorithmic fairness and data leakage**: quantified per-sex recall disparity (0.985 for men vs. 0.846 for women; 0.139 gap logged to `metrics.json` and documented in the model card) and eliminated a pre-split missingness feature-selection leakage path.
- **Productionized into a modular `src/` package** with FastAPI, Docker, and automated test suites, verifying strict train/serve parity across all 230 test rows (catching and fixing three silent skew bugs) at **60–80 ms p95 latency on GCP Cloud Run**.
- **Shipped the complete MLOps lifecycle**: DVC data versioning, MLflow model registry with `@champion` release gating, GitHub Actions + CML CI blocking PRs that drop test AUC > 0.02, automated weekly PSI drift monitoring, and a public Streamlit explainability UI.

Home Credit Default Risk – Large-Scale Gradient Boosting Architecture [GitHub] XGBoost • LightGBM • CatBoost • Optuna

- **Engineered an end-to-end credit-default prediction pipeline** across seven relational source tables (**307,511 borrowers**, 8.07% default rate), constructing **666 domain and aggregation features** behind an automated data-audit sweep for disguised missing values and sentinel codes.
- **Controlled regularization across all three GBDT challengers** (L2 = 5.0, 0.85 row and 0.5 column subsampling, 200-round early stopping) to isolate inductive bias from hyperparameter tuning; rank-averaged ensemble reached **0.79566 private-leaderboard ROC-AUC** vs. 0.79440 for Optuna-tuned XGBoost.
- **Quantified cross-validation variance**: fold-score spread (0.010) ran 5.7× the 0.0018 gap between best and worst model, relying strictly on pooled out-of-fold AUC to predict the held-out private leaderboard within **0.0001**.

> **Additional Deep Learning Implementations:** **Vision Transformer (ViT) from Scratch in PyTorch** (patch embeddings, multi-head self-attention, and custom training loop).

EDUCATION

Panjab University, SSGRC Hoshiarpur — B.E. in Computer Science & Engineering **2014 – 2018**

SSDPS Hoshiarpur — Senior Secondary (PCM) **2013**